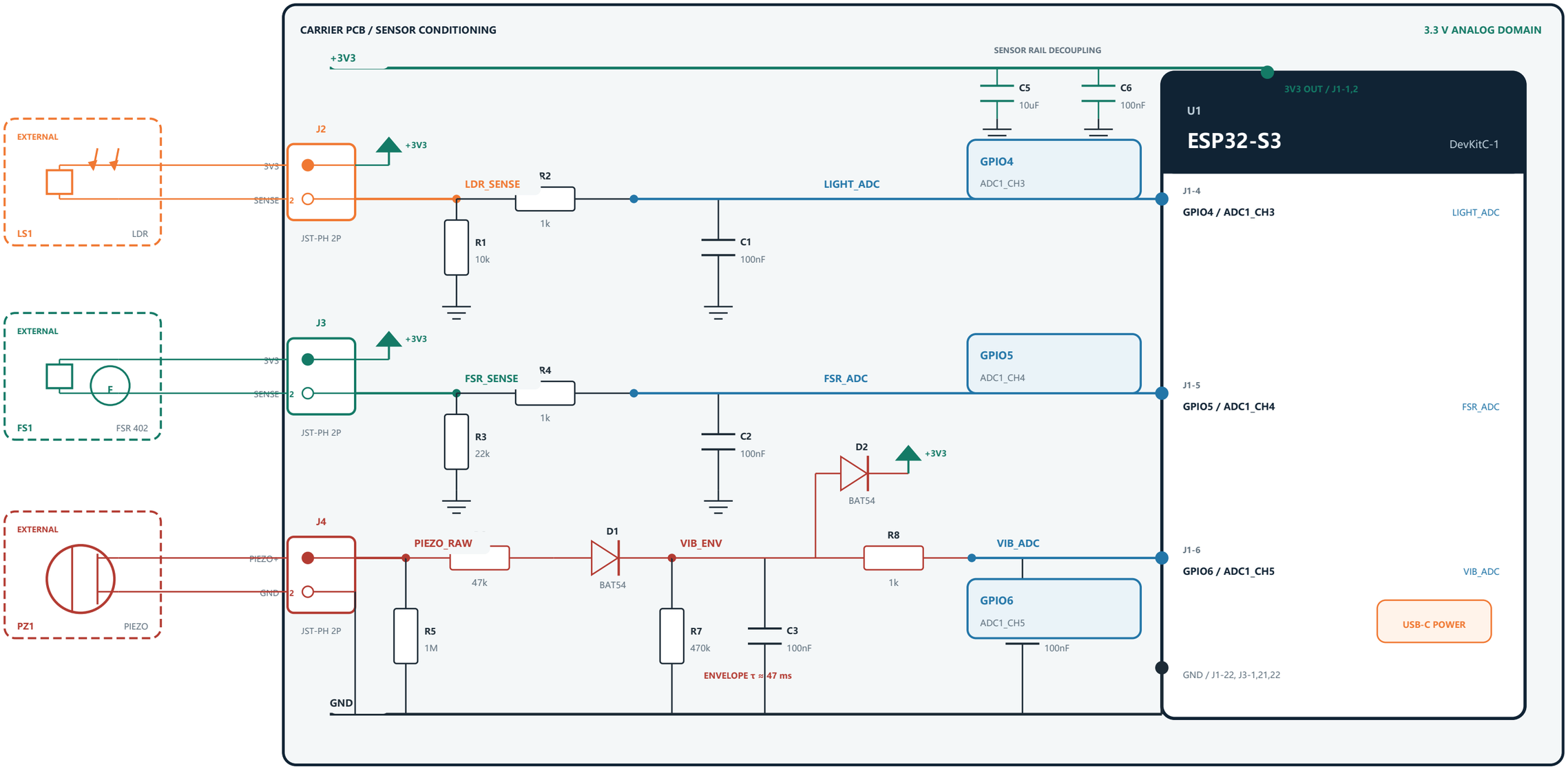
ESP32-S3 SENSOR FRONT-END

AMBIENT LIGHT · THIN-FILM PRESSURE · VIBRATION ENVELOPE



DESIGN INTENT

01 ADC1 only

GPIO4 / GPIO5 / GPIO6 avoid USB D-/D+ and follow Espressif's ADC1 preference.

02 Protected piezo

R6 limits current; D1 rectifies; C3/R7 hold the peak; D2 clamps to the 3V3 rail.

03 Human calibration

LDR and FSR thresholds must be learned from the installed enclosure and real users.

04 Keyed harnesses

JST-PH 2P, square-pad Pin 1, fixed wire colors, common 3V3/GND domain.